

SPECIAL CASTING PROCESSES

Shell Molding

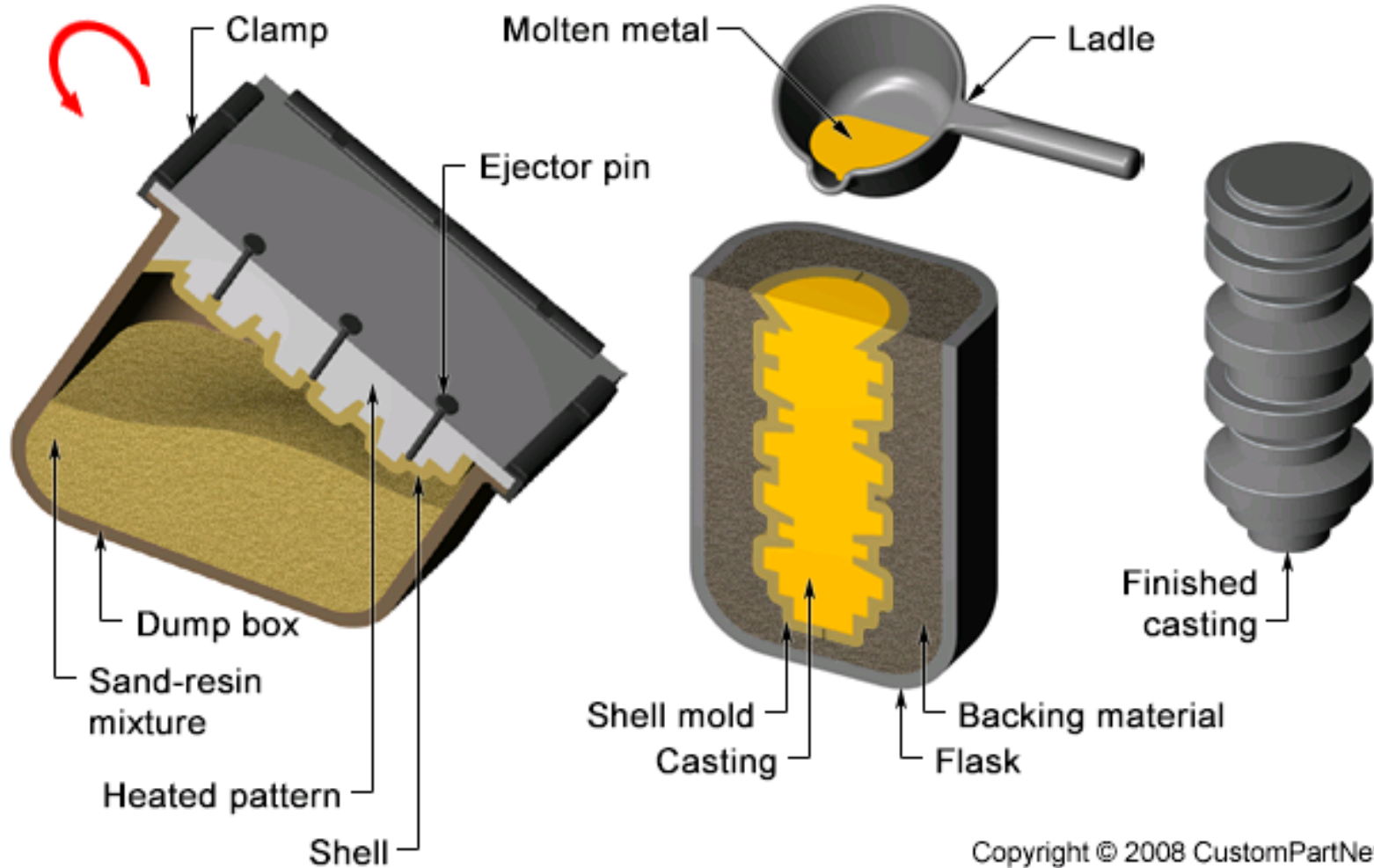
Shell molding is a process for producing simple or complex near net shape castings, maintaining tight tolerances and a high degree of dimensional stability.

1. High quality castings.
2. Can be manufactured for wider range of alloys
3. At a lower cost than investment casting.

Shell-Making (Cross-section)

Shell Mold Casting (Cross-section)

Casting



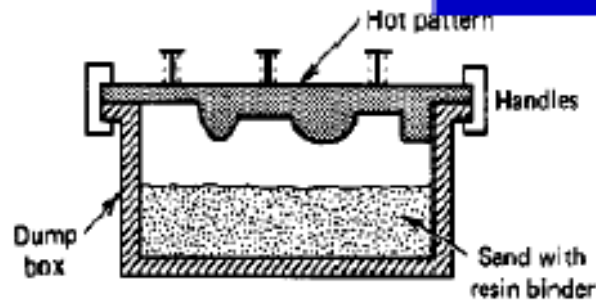
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- First a pattern is made of iron. The iron pattern is heated to $\pm 200^{\circ}\text{C}$.
- The pattern is attached to a dump box and sprayed with a releasing agent; similar to putting oil in a baking pan.
- The dump box contains fine sand mixed with thermo setting resin binder.
- The dump box is rotated with the pattern now at the bottom and the sand that was once at the bottom is now on top.
- The sand mixture falls against the pattern and gets heated in this manner, the resin cures, causing the sand grains to adhere to each other forming sturdy shell that exactly conforms to the dimensions and shape of the pattern and constitutes half of a mold.
- After a 5mm thick shell is formed, due to melting and solidifying of the thermosetting binder, the dump box is rotated and the unused sand falls to the bottom.
- The pattern and shell are then put in an oven at 350°C – 400°C to cure the resin.
- The pattern is removed by the ejector pins.
- Typically, two pattern shells are used together. A backing material like sand or steel shot is used and the metal is poured into the mould.

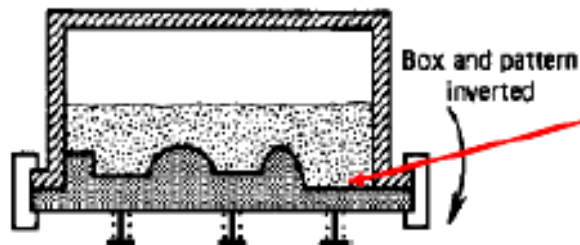
Raw Materials, Equipments for shell Mould Casting

- Resin coated sand.
- Chromel Alumel Thermocouple, digital panel meter.
- Muffle Furnace.
- 2 Kg Aluminium Silicon Alloy (LM6) scrap.
- Clay Graphite crucible.
- Fireclay.
- Asbestos Gloves, Tongs, safety goggles.

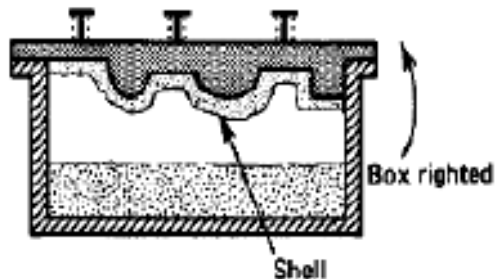
The shell moulding process



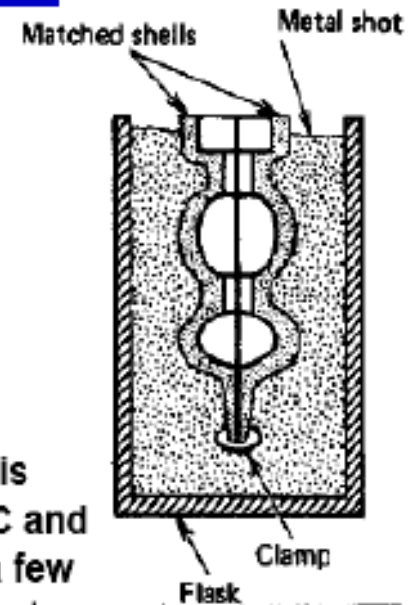
(a)



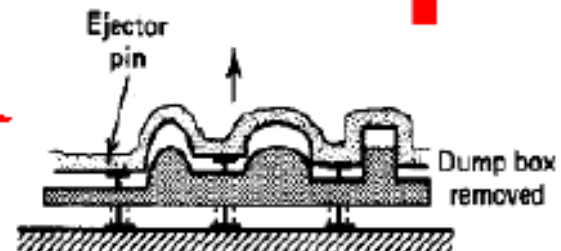
(b)



(b) The metal pattern is heated to 150 – 232 °C and allowed to stand for a few minutes, while the sand-plastic mixture forms a 5mm thick, solid layer.



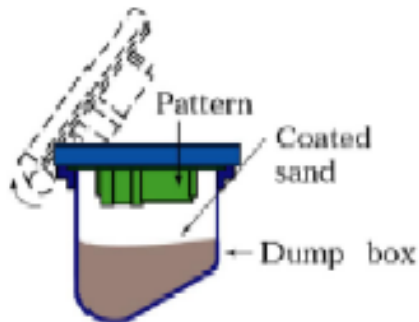
(e)



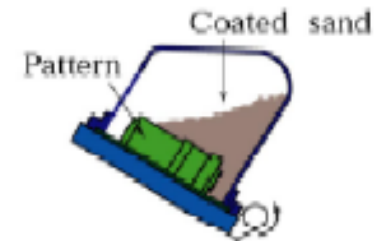
- Developed in the 1940's
- Produces close dimensional tolerances
- Good surface finish
- Low cost process

Common methods of making shell molds.

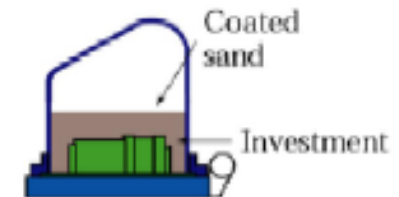
(a) Pattern rotated and clamped



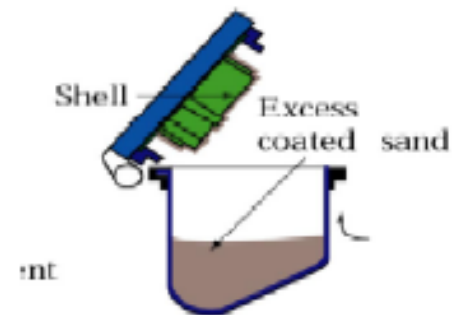
(b) Pattern and dump box rotated

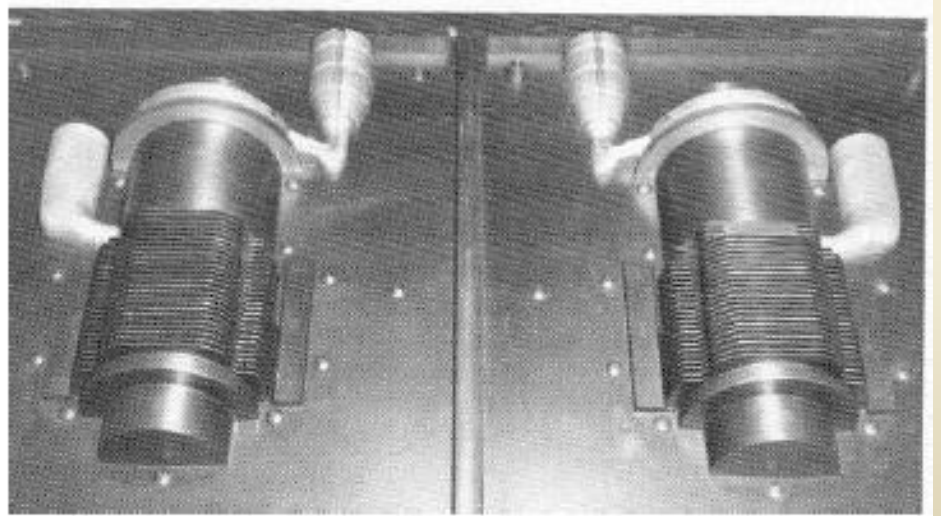
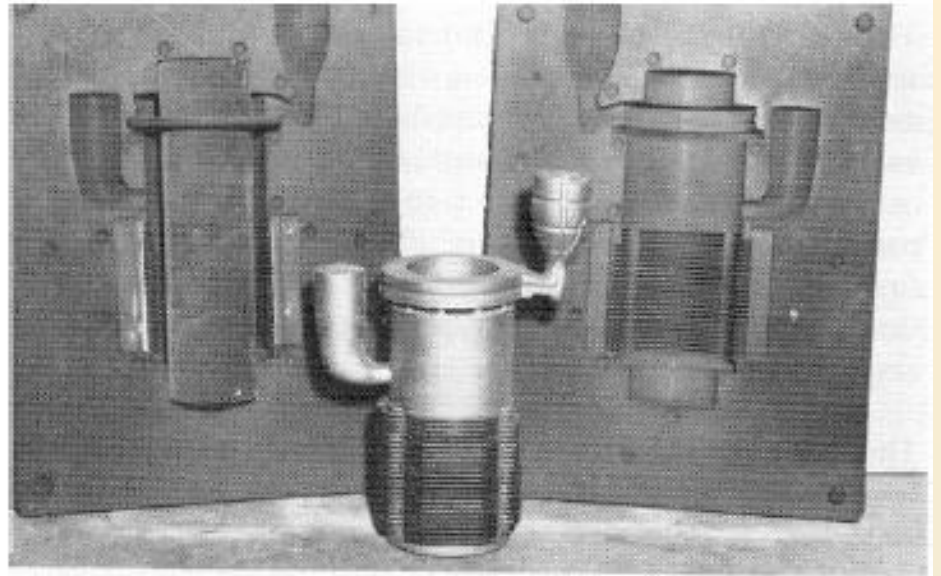
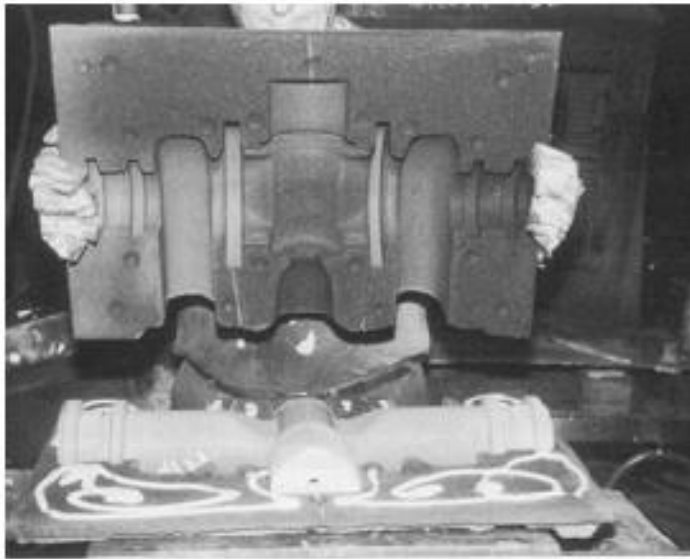


(c) Pattern dump box in position for the investment



(d) Pattern and shell removed from dump box





Figures showing two different shell molds

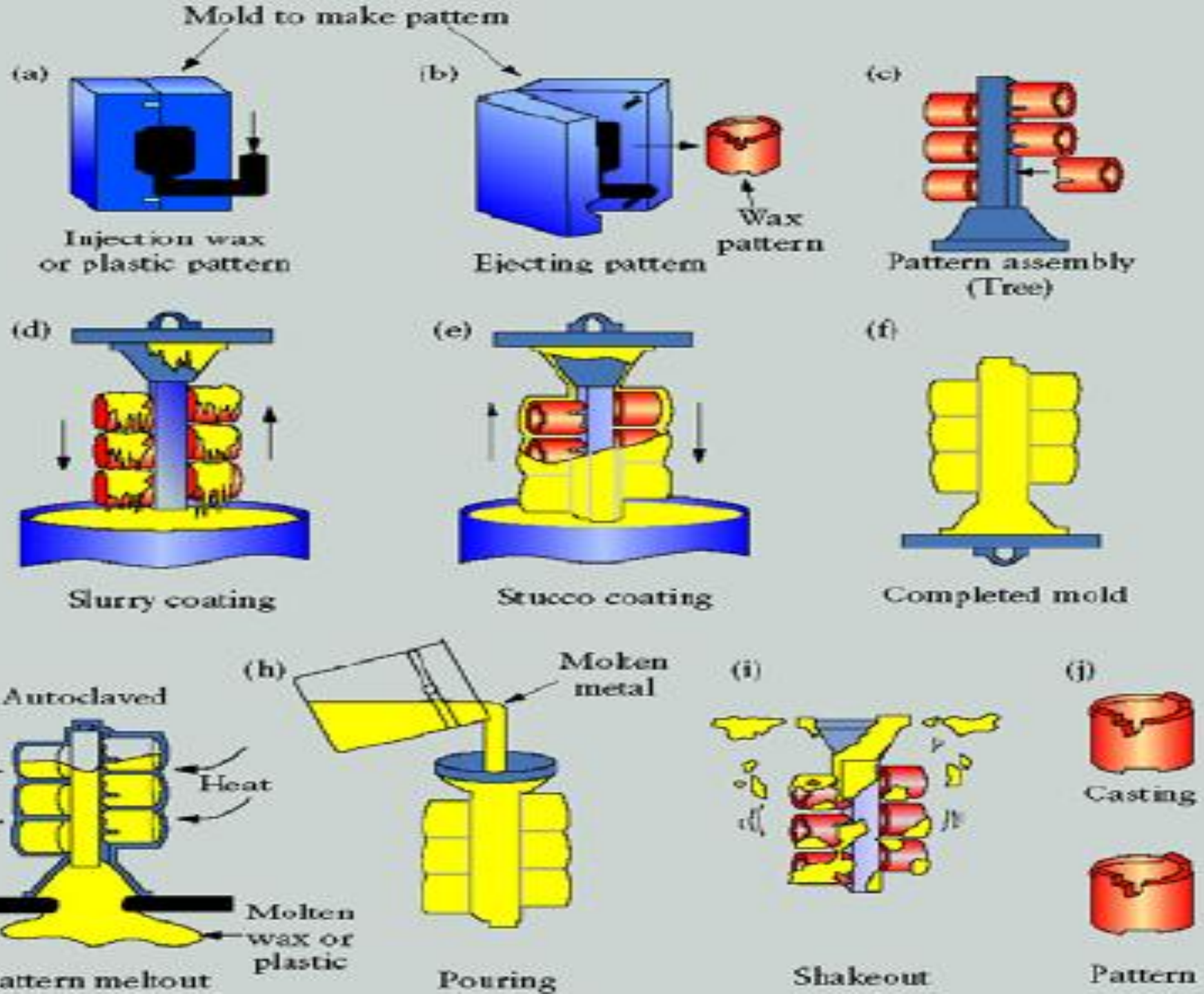
- **Advantages:**
 - Better surface finishes are obtained.
 - Machining of castings are reduced.
 - The chances of blow holes or pockets are reduced since the holes are highly permeable.
 - Shells can be stored for long time.
 - High dimensional accuracy can be obtained.
- **Limitations:**
 - The metal pattern is costlier than the wooden one.
 - High resin cost.
 - High equipment cost.
 - Uneconomical for small production lots.
 - Relative inflexibilities in gating and risering systems.
- **Applications:**
 - The most common application is for automobile parts castings in grey cast iron and aluminum.

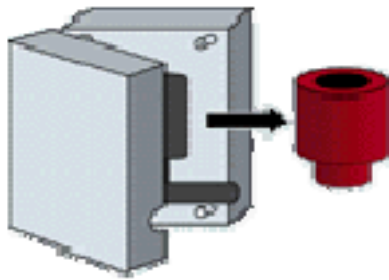
Investment Casting (Lost Wax Process)

- **Investment casting** is also known as the lost wax process.
- This process is one of the oldest manufacturing processes.
- The Egyptians used it in the time of the Pharaohs to make gold jewelry (hence the name Investment) some 5,000 years ago.
- Intricate shapes can be made with high accuracy.
- In addition, metals that are hard to machine or fabricate are good candidates for this process.
- It can be used to make parts that cannot be produced by normal manufacturing techniques, such as turbine blades that have complex shapes, or airplane parts that have to withstand high temperatures.
- The mold is made by making a pattern using wax or some other material that can be melted away.
- This wax pattern is dipped in refractory slurry, which coats the wax pattern and forms a skin.
- This is dried and the process of dipping in the slurry and drying is repeated until a robust thickness is achieved.

- After this, the entire pattern is placed in an oven and the wax is melted away.
- This leads to a mold that can be filled with the molten metal.
- Because the mold is formed around a one-piece pattern, (which does not have to be pulled out from the mold as in a traditional sand casting process), very intricate parts and undercuts can be made.
- The materials used for the slurry are a mixture of plaster of Paris, a binder and powdered silica, a refractory, for low temperature melts.
- For higher temperature melts, sillimanite an alumina-silicate is used as a refractory, and silica is used as a binder.
- Depending on the fineness of the finish desired additional coatings of sillimanite and ethyl silicate may be applied.
- The mold thus produced can be used directly for light castings, or be reinforced by placing it in a larger container and reinforcing it more slurry.
- Just before the pour, the mold is pre-heated to about 1000 °C (1832 °F) to remove any residues of wax, harden the binder.

- The pour in the pre-heated mold also ensures that the mold will fill completely.
- Pouring can be done using gravity, pressure or vacuum conditions.
- Attention must be paid to mold permeability when using pressure, to allow the air to escape as the pour is done.
- The types of materials that can be cast are Aluminum alloys, Bronzes, tool steels, stainless steels, Stellite, and precious metals.
- Parts made with investment castings often do not require any further machining, because of the close tolerances that can be achieved.



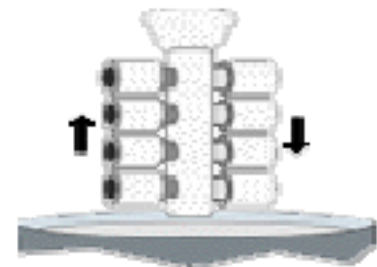


1. WAX INJECTION : Wax replicas of the desired castings are produced by injection molding. These replicas are called patterns.



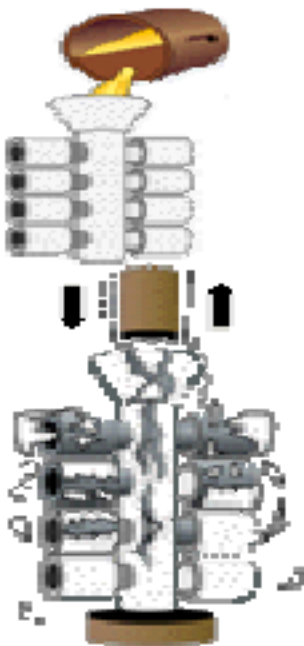
2. ASSEMBLY : The patterns are attached to a central wax stick, called a sprue, to form a casting cluster or assembly.

3. SHELL BUILDING : The shell is built by immersing the assembly in a liquid ceramic slurry and then into a bed of extremely fine sand. Up to eight layers may be applied in this manner.



4. DEWAX : Once the ceramic is dry, the wax is melted out, creating a negative impression of the assembly within the shell.

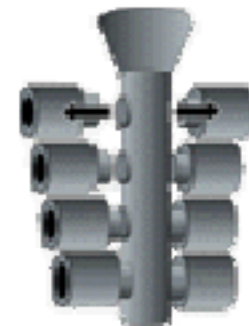




- **5. CONVENTIONAL CASTING**
In the conventional process, the shell is filled with molten metal by gravity pouring. As the metal cools, the parts and gates, sprue and pouring cup become one solid casting.

- **6. KNOCKOUT**
When the metal has cooled and solidified, the ceramic shell is broken off by vibration or water blasting.

- **7. CUT OFF**
The parts are cut away from the central sprue using a high speed friction saw.

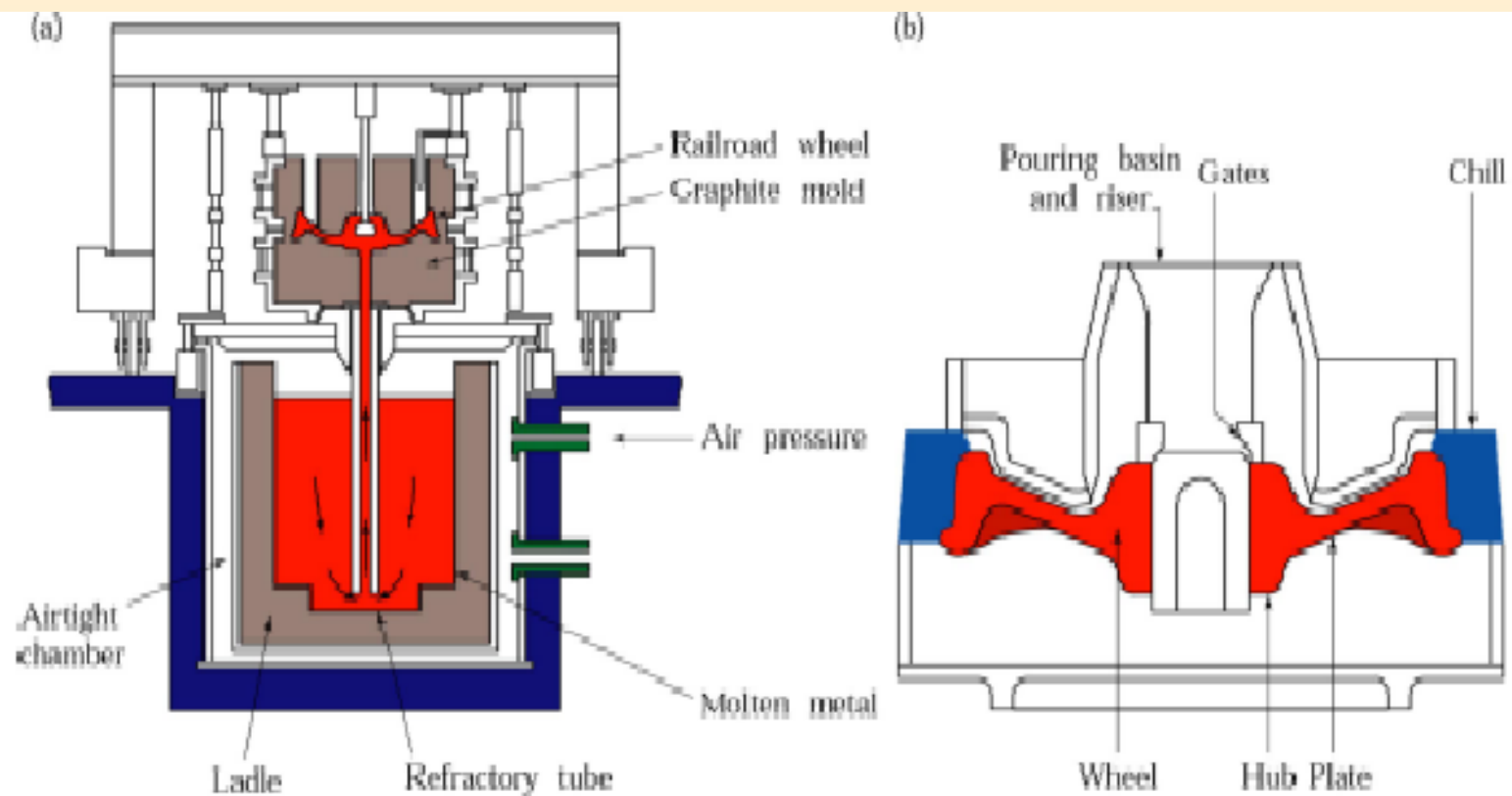


- **8. FINISHED CASTINGS**
After minor finishing operations, the metal castings--identical to the original wax patterns--are ready for shipment to the customer.



- **Advantages:**
 - High dimensional accuracy.
 - A very smooth surface can be achieved.
 - Extremely thin sections, to the extent 0.75mm, can be cast successfully.
 - Suitable for mass production of small sized castings.
- **Limitations:**
 - Unsuitable for castings of more than about 5 Kg.
 - The process is expensive.
 - Skilled worker is required.
 - Production rate is slow.
- **Applications:**
 - This process is used for casting of turbine blades and parts of automobile.

Pressure Casting



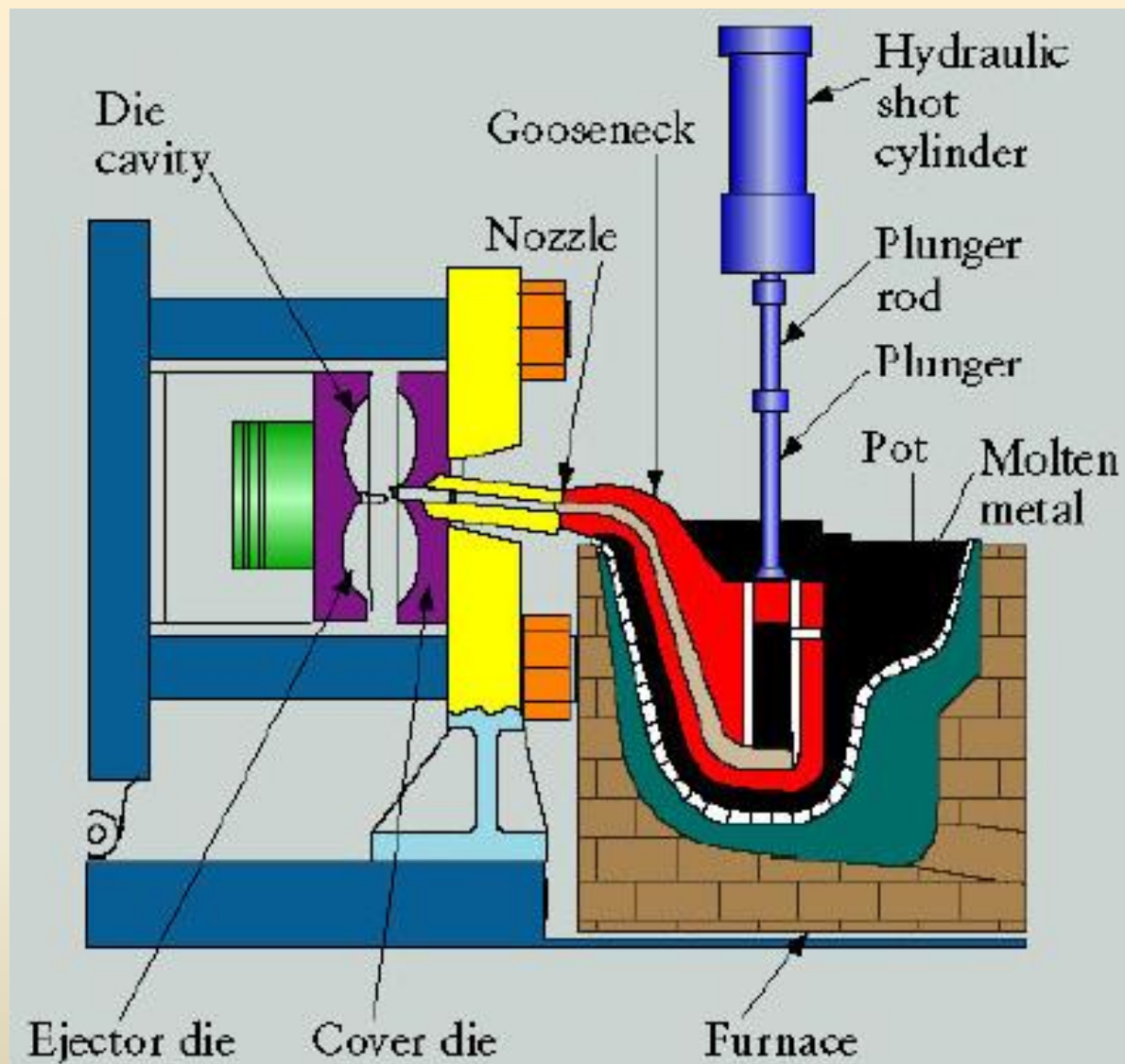
(a) The bottom-pressure casting process utilizes graphite molds for the production of steel railroad wheels. (b) Gravity pouring method of casting a railroad wheel. Note that the pouring basin also serves as a riser.

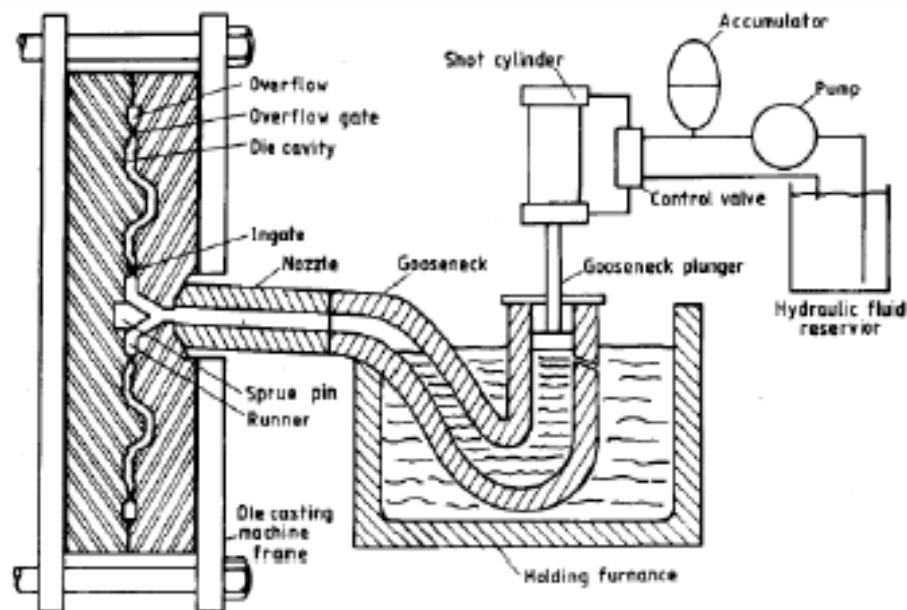
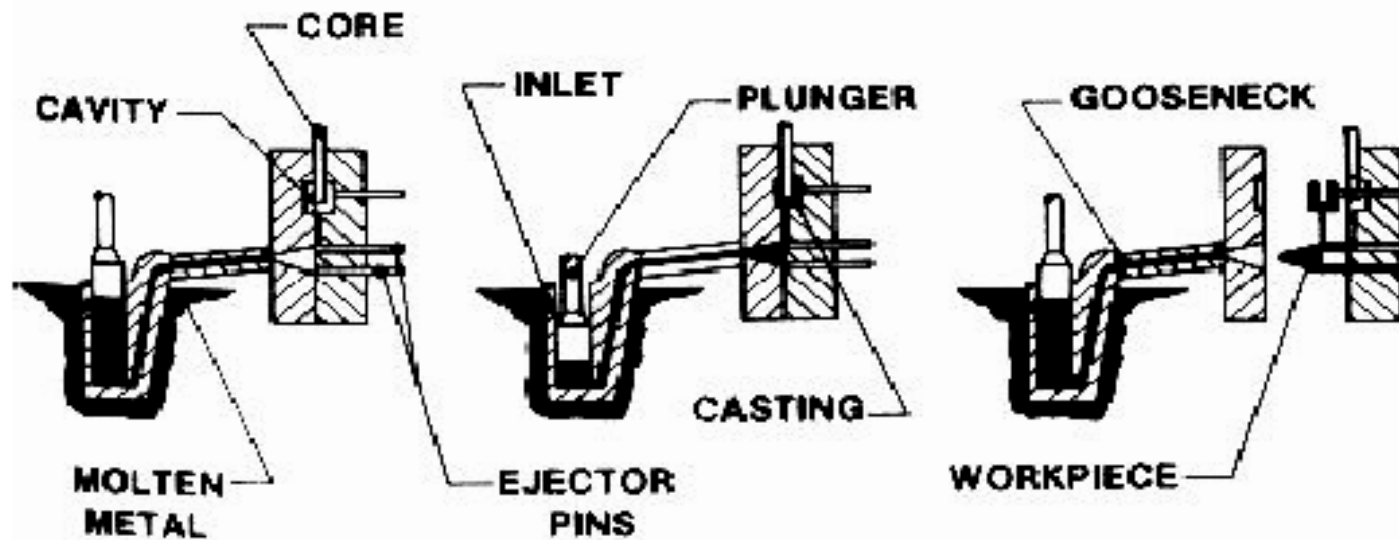
- Used a graphite or metal mold
- Molten metal is forced into the mold by gas pressure
- The pressure is maintained until the metal solidifies in the mold
- Used for high-quality castings

Die Casting (Multi-use Mold Casting)

- The term die is used for permanent mold.
- In this process the molten metal is forced into the permanent mold (dies) under high pressure.
- The molten metal fills the entire die, including the minute details.
- On solidification the casting is taken out.
- Thus High pressure die casting uses a piston to inject the molten metal into the die.
- This greatly speeds the process, and therefore increases production.
- This results in a more uniform part, generally good surface finish and good dimensional accuracy, as good as 0.2 % of casting dimension.
- For many parts, post-machining can be totally eliminated, or very light machining may be required to bring dimensions to size.
- There are two types of high pressure die casting: *hot chamber* and *cold chamber*.

- **Hot chamber die casting:**
- The melting unit of the metal forms an integral part of the machine.
- The plunger is made up of refractory material.
- When the plunger is raised, it uncovers an opening in the cylinder wall, through which molten metal enters, filling the cylinder.
- The molten metal is forced into the die either by hydraulic pressure or by air pressure applied to the plunger.
- As soon as the metal solidifies, the pressure on the metal is relieved and the plunger travels upwards to its original position.
- The casting is ejected from the die by means of ejector pins.
- This process is particularly suitable for lead, tin and zinc alloys.
- Hot chamber die casting cannot be used for metals having high melting temperatures.
- Also it can be used to cast metals like aluminum, copper etc. which will react with steel at high temperature.

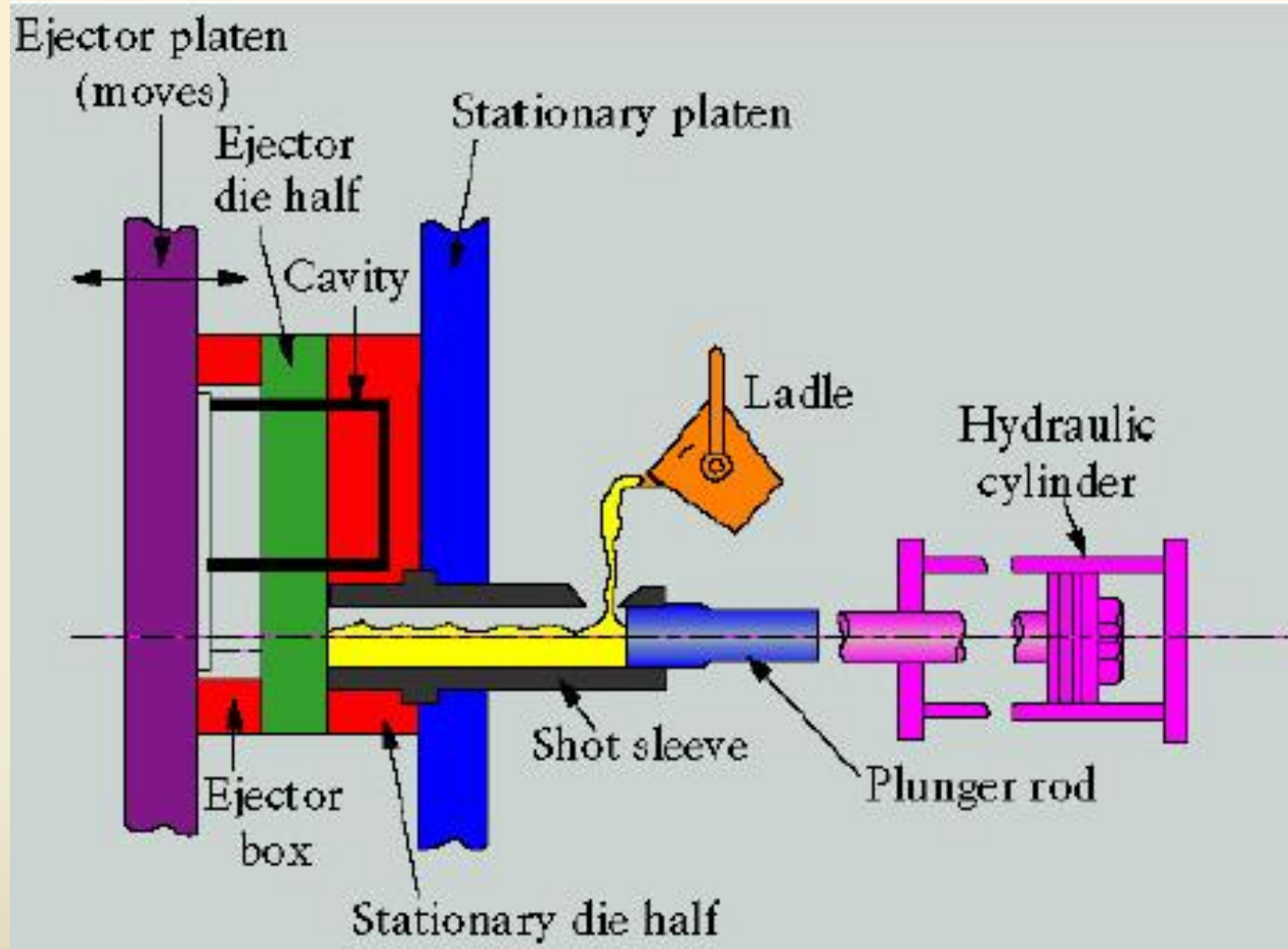


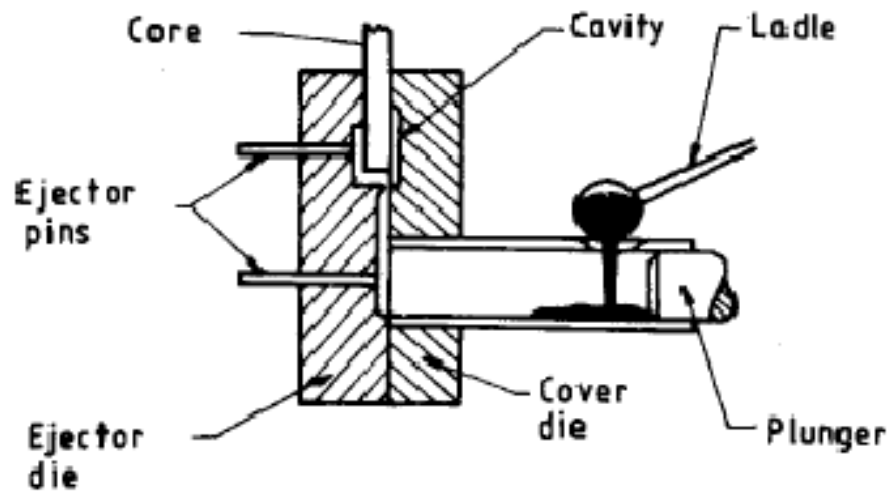


Hot chamber die casting

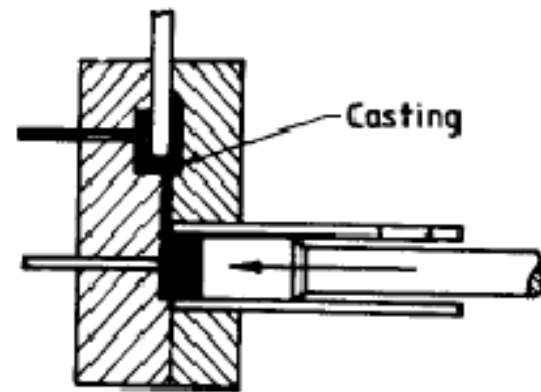
- **Cold chamber die casting:**

- In cold chamber die casting the plunger is horizontal and is driven by air or hydraulic pressure to force the molten metal in to the die.
- As soon as the ladle is emptied, plunger moves forward and forces the metal into the cavity of the die.
- After the metal solidifies, the core is with drawn, and then the die is opened.
- Ejector pins are employed to remove the casting automatically from the die.
- The cold chamber die casting is particularly suitable for metal such as aluminum alloys, magnesium, copper, which cannot be cast in hot chamber machines due to reactivity with molten aluminum and steel.
- High melting temperature alloys of non-ferrous type are best die cast in cold chamber die casting.

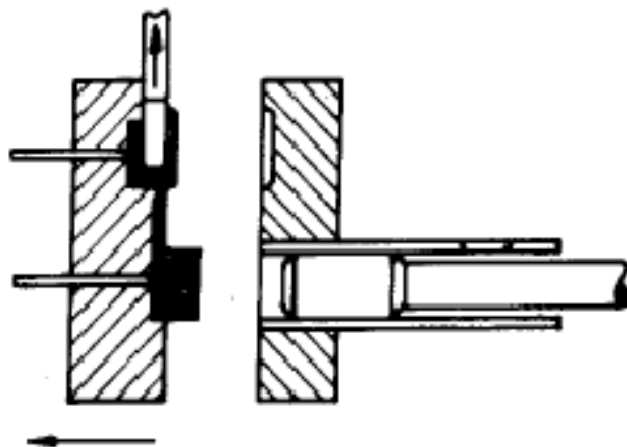




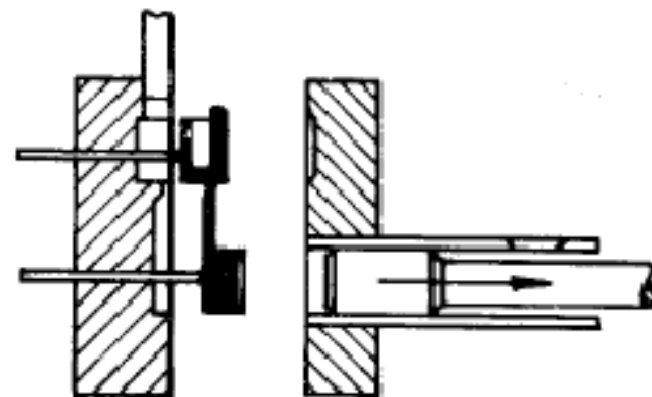
(a)



(b)



(c)



(d)

Advantages of die casting:

- Very high production rates are possible. Typical rate could be 200 pieces per hour since the process is completely automated.
- Thin sections can be cast. Very small thicknesses can be easily filled because the liquid metal is injected at high pressure. Surface of about 0.8 microns can be achieved.
- Close dimensional tolerances of the order of $\pm 0.0025\text{mm}$ is possible.
- Because of the use of the movable cores, it is possible to obtain fairly complex castings.
- Fine details may be produced.
- Less floor space is required.

Limitations of die casting:

- Not economical for small runs. The dies and the machines are very expensive and therefore, economy in production is possible only when large quantities are produced.
- Only economical for non-ferrous alloys.
- Due to high temperature of molten metal, die life decreases.
- The air in the die cavity gets trapped inside the casting and is therefore a problem often with the diecastings.
- The maximum size of the casting is limited. The normal sizes are under 4 kg with a maximum of the order of 15 kg because of the limitation on the machine capacity.

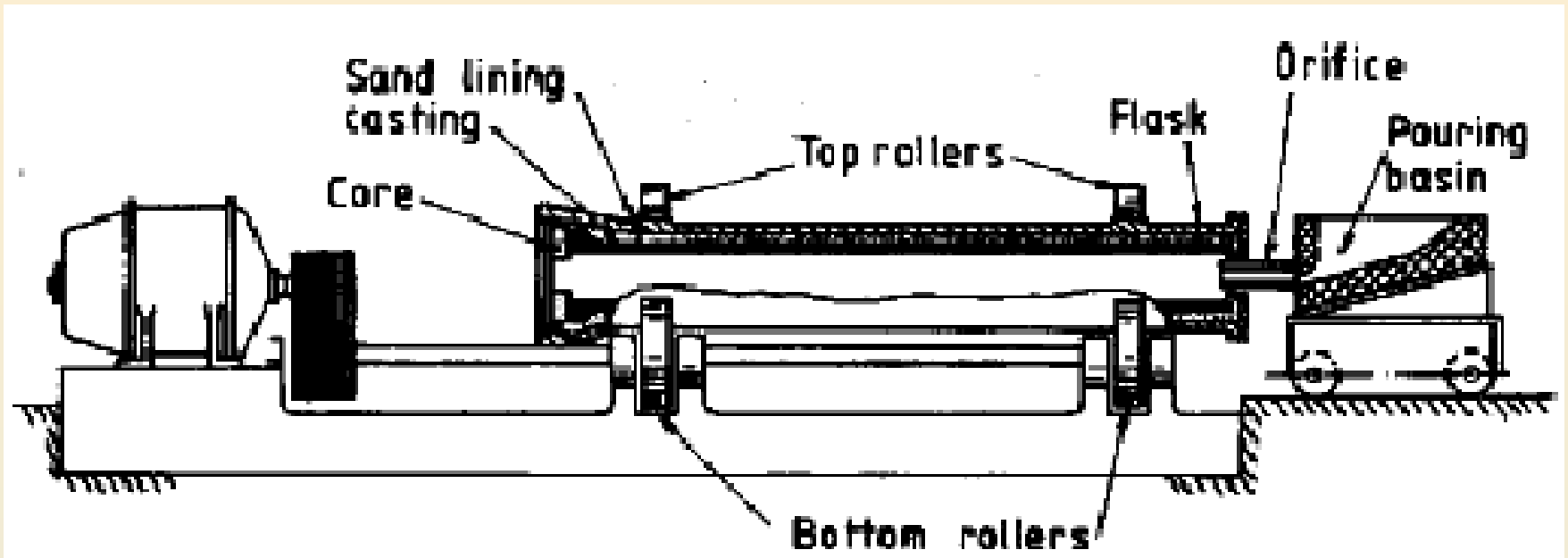
Centrifugal Casting

Centrifugal casting are of three types:

- ***True centrifugal casting:*** Long molds are rotated about a horizontal axis. This can be used to make long axial parts such as seamless pipes.
- ***Semi centrifugal casting:*** Parts with a wide radial parts. Parts such as wheels with spokes can be made with this technique
- **Centrifuging:** The molds are placed a distance from the center of rotation. Thus when the poured metal reaches the molds there is a high pressure available to completely fill the cavities. The distance from the axis of rotation can be increased to change the properties

True Centrifugal Casting

- A mold is set up and rotated along a inclined or horizontal (200-1000 rpm) axis.
- The mold is coated with a refractory coating.
- While rotating molten metal is poured in.
- The metal that is poured in will then distribute itself over the rotating wall.
- During cooling lower density impurities will tend to rise towards the center of rotation.
- After the part has solidified, it is removed and finished.



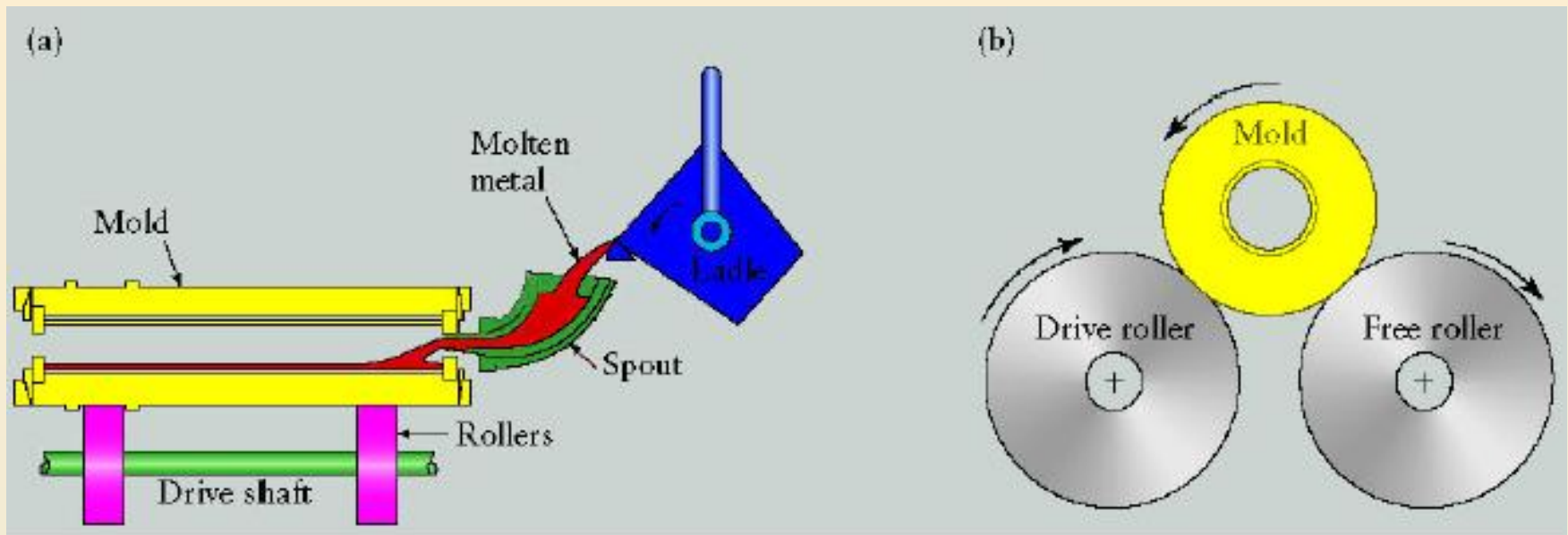


FIGURE: Schematic illustration of the centrifugal casting process. Pipes, cylinder liners, and similarly shaped parts can be cast by this process.

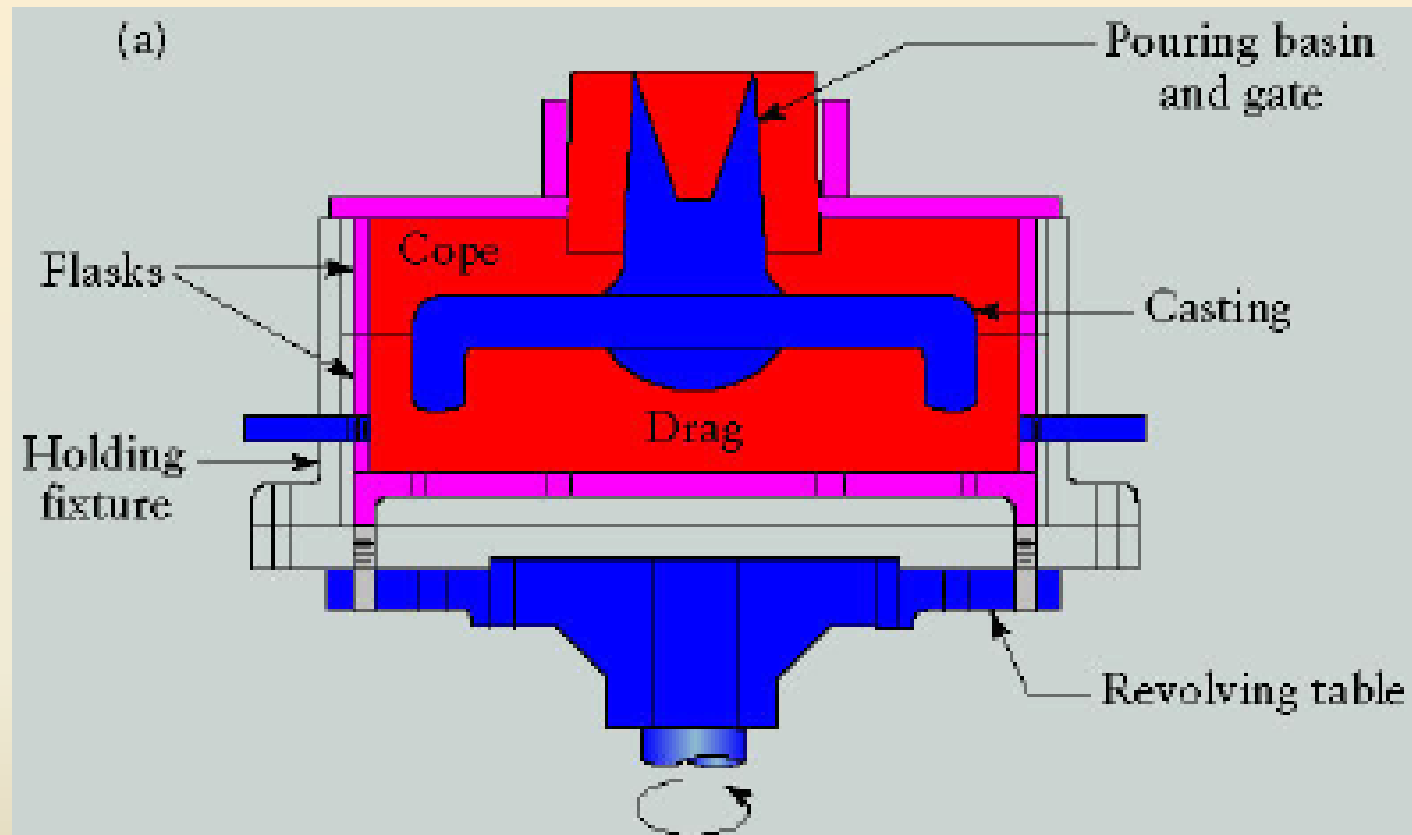
Advantages:

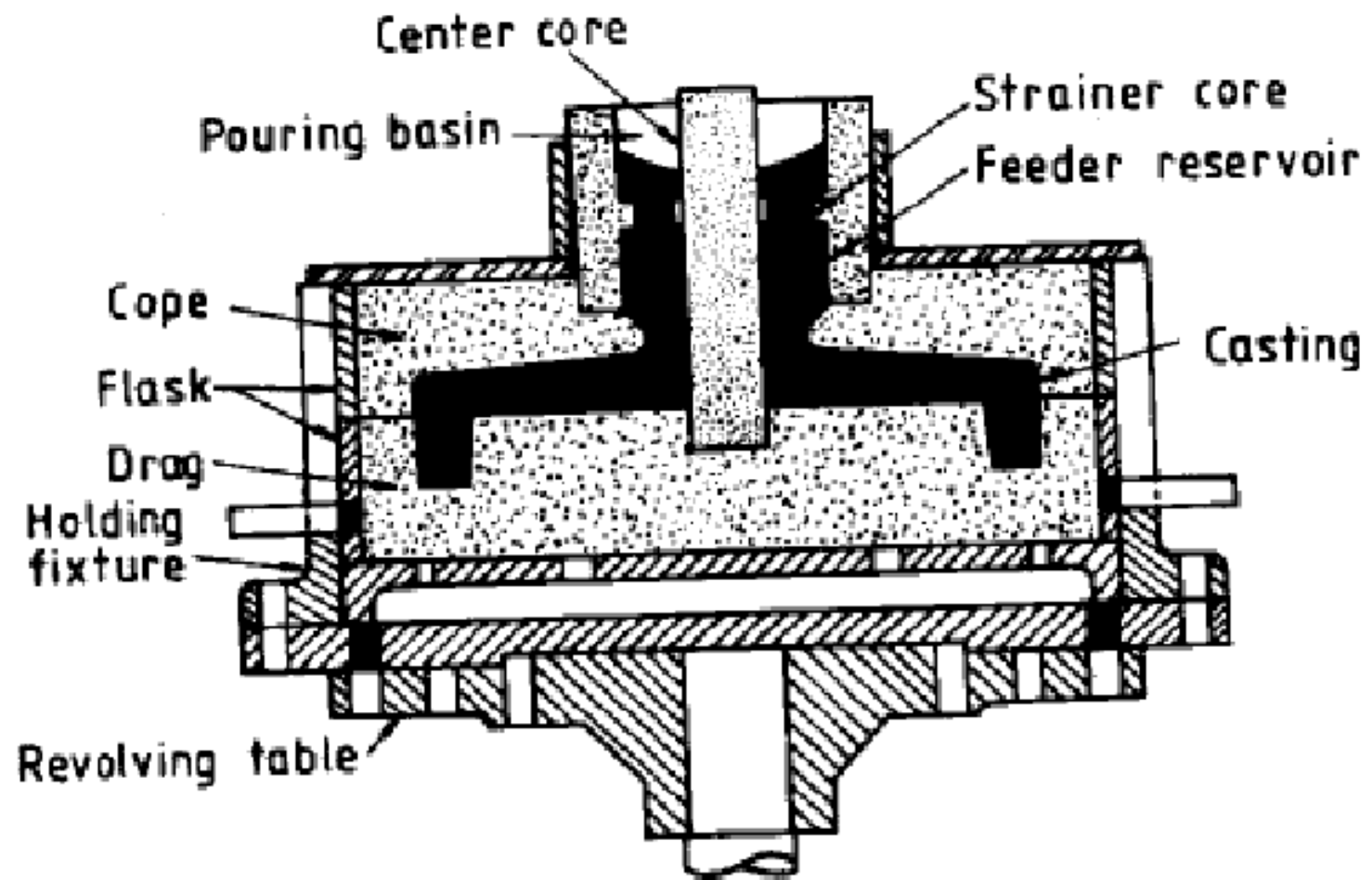
- The mechanical properties of centrifugally cast jobs are better compared to other processes, because the inclusions such as slag and oxides gets segregated towards the centre and can be easily removed by machining. Also, the pressure acting on the metal throughout the solidification, causes the porosity to be eliminated giving rise to dense metal.
- Good uniform metal properties
- No sprues/gates to remove
- The outside of the casting is at the required dimensions
- Lower material usage
- No cores are required for making concentric holes in the case of true centrifugal casting.

Disadvantages:

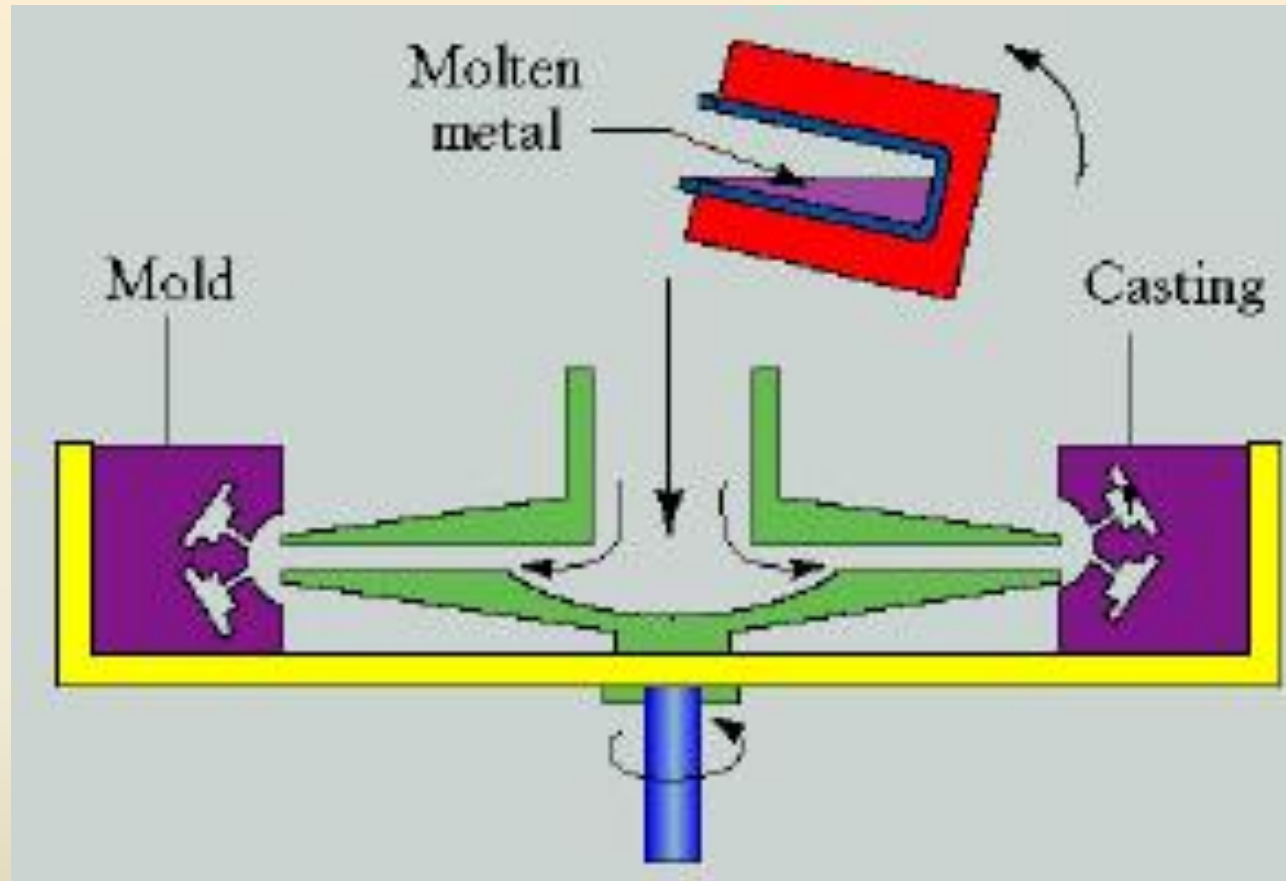
- Extra equipment needed to spin mold
- The inner metal of the part contains impurities .
- Only certain shapes which are axi-symmetric and having concentric holes are suitable for true centrifugal casting.
- The equipment is expensive and thus is suitable only for large quantity production.

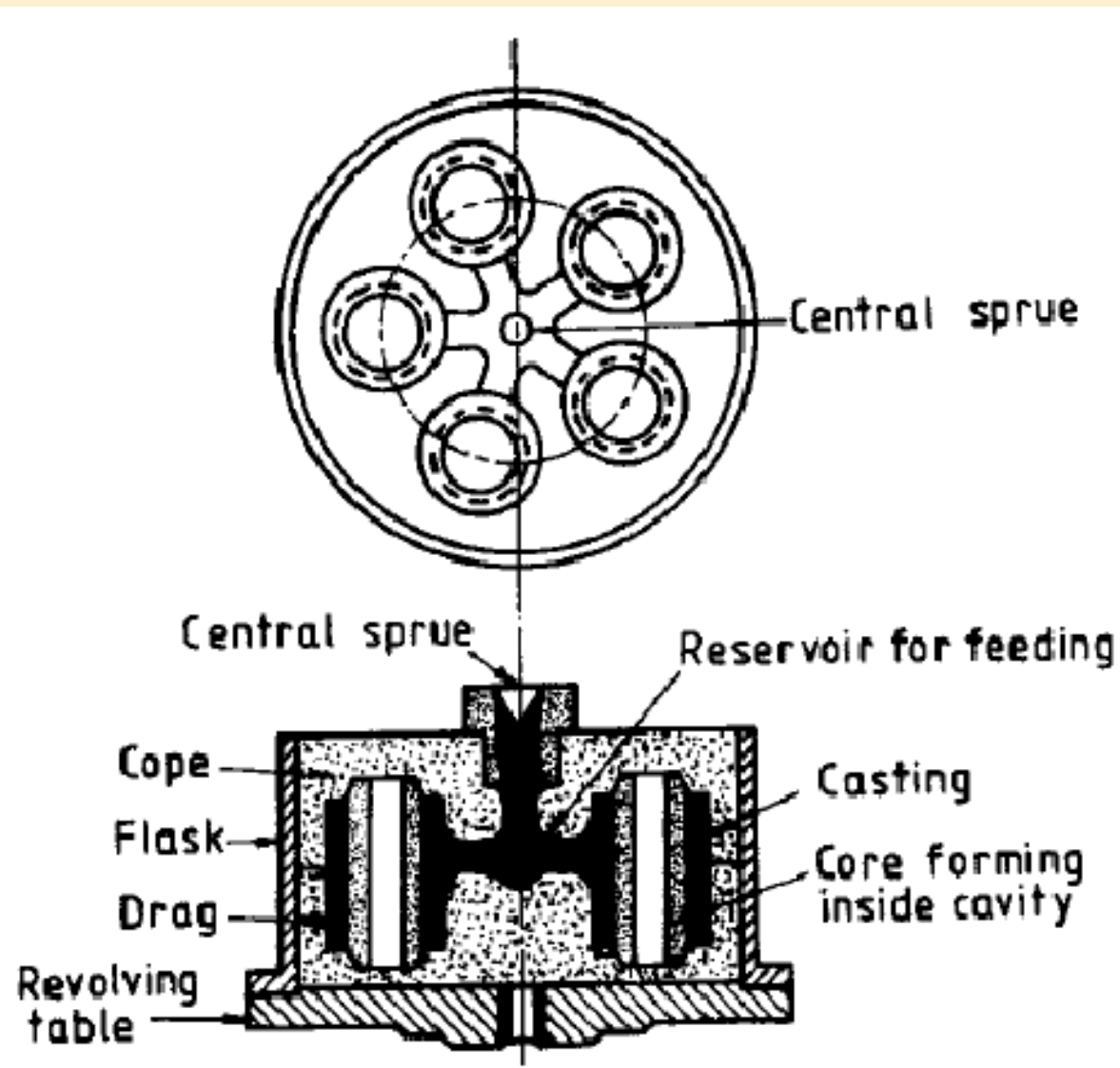
Semi-Centrifugal Casting





Centrifuging Casting





Casting Defects

Several types of defects may occur during casting, considerably reducing the total output of castings besides increasing the cost of their production. It is therefore essential to understand the causes behind these defects so that they may be suitably eliminated.

Casting defects may be defined as those characteristics that create a imperfection or deficiency contrary to the quality specifications imposed by the design and service requirements.

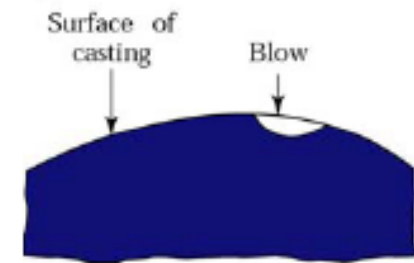
The most common casting defects can be classified into the following categories:

- Gas defects
- Shrinkage cavities
- Molding material defects
- Pouring metal defects
- Metallurgical defects
- Mold preparation defects.

Gas Defects

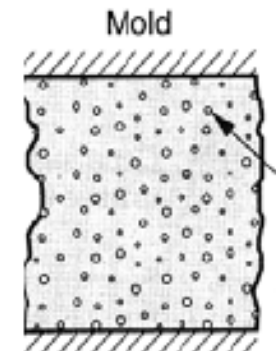
Blow holes and open blows:

- These are the spherical, flattened or elongated cavities present inside the casting or on the surface.
- On the surface they are called open blows and inside, they are called blow holes.
- It is an excessively smooth depression on the outer surface of a casting. Blow holes are entrapped bubbles of gases with smooth walls.
- *Causes:*
 - High moisture content in the molding sand.
 - Low permeability of molding sand.
 - Hard ramming of the sand.
 - Improper venting.
- *Remedies:*
 - This defect can be eliminated by ensuring proper venting and using proper molding sand.



Air inclusions or Gas Holes or Internal air pocket

- The atmospheric and other gases absorbed by the molten metal in the furnace, in the ladle, and during the flow in the mould, when not allowed to escape, would be trapped inside the casting and weaken it.
- This appears as small holes inside the casting and is caused by rapid pouring of the molten metal.
- *Causes:*
 - Higher pouring temperature which increases the amount of gas absorbed.
 - Poor quality of metal.
 - Excessively moist sand.
- *Remedies:*
 - This defect can be eliminated by ensuring proper pouring temperature, and by degasifying the molten metal.



Pin hole porosity:

- These are numerous holes of small diameter, usually less than 2mm, visible on the outer surface of the casting.
- These are caused by the absorption of hydrogen or carbon monoxide when the moisture content of the sand is high.
- Causes:
 - High moisture content in the molding sand.
 - Low permeability of molding sand.
 - Improper melting and fluxing practices.
- Remedies:
 - This defect can be eliminated by using proper molding sand and using proper melting and fluxing practice.

Shrinkage cavities:

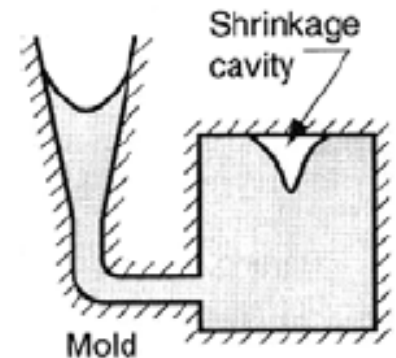
- These are caused by the liquid shrinkage occurring during the solidification of the casting.
- It is void or depression in the casting caused mainly by uncontrolled and haphazard solidification of metal.
- To compensate this, proper feeding of liquid metal is required as also proper casting design.

- *Causes:*

- Incorrect metal composition.
- Incorrect pouring temperature.
- Improper location and size of gates and runner.

- *Remedies:*

- This defect can be eliminated by ensuring proper mold design, so that solidification rate is uniform.



Molding Material Defects

Cuts and washes or Scabs:

- These appear as rough spots and areas of excess metal, and are caused by the erosion of molding sand by the flowing molten metal.
- This may be caused by the molding sand not having enough strength or the molten metal flowing at high velocity.
- Scabs are sort of projection on the casting which occur when a portion of the mold face or core lifts and the metal flows beneath in thin layer.
- During casting when sand face of the mold gets heated up and expands, sometimes there occurs a crack.
- Molten metal enters the crack and flows behind the layer of sand, and causes scabs.
- *Causes:*
 - Using molding sand of low strength.
 - Using very fine sand having low permeability and moisture content.
- *Remedies:*
 - This defect can be eliminated by mixing additives such as wood flour, sea coal.



Metal penetration:

- When the molten metal enters the gaps between the sand grains, the result would be a rough casting surface.
- It occurs when molten metal being cast tends to penetrate into the sand grains and causes a fused aggregate of metal and sand on the surface of the casting.
- *Causes:*
 - Improper ramming of sand.
 - Excessive pouring temperature which increases the fluidity of metal.
- *Remedies:*
 - This defect can be eliminated by fine sand and hard ramming the sand.

Fusion:

- This is caused by the fusion of sand grains with the molten metal, giving a brittle, glassy appearance on the casting surface.
- *Causes:*
 - Using molding sand of low strength.
 - Improper ramming of sand.
- *Remedies:*
 - This defect can be eliminated by sand of proper strength and the sand should be properly rammed in.

Runout :

- A run out is caused when the molten metal leaks out of the mould.
- *Causes:*
 - This may be caused either due to faulty mould making or because of the faulty molding flask.

Drop:

- The dropping of loose molding sand or lumps normally from the cope surface into the mould cavity is responsible for this defect.
- This is essentially, due to improper ramming of the cope flask.
- Sometimes during casting upper surface of the mold cracks and pieces of sands fall into the molten metal, this is known as drop.
- *Causes:*
 - Using molding sand of low strength.
 - Improper ramming of sand.
- *Remedies:*
 - This defect can be eliminated by sand of proper strength and the sand should be properly rammed in.

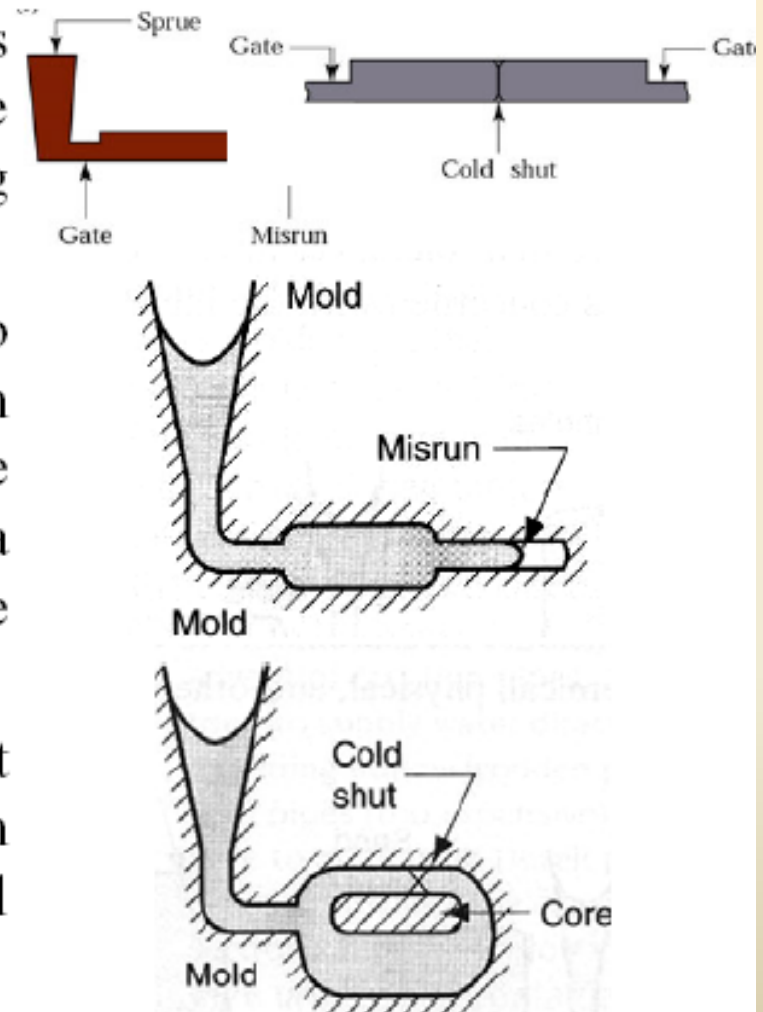
Swell:

- Under the influence of the metallostatic forces, the mould wall may move back causing a swell in the dimensions of the casting.
- It is a localized enlargement of the casting due localized enlargement of the mold by molten metal pressure.
- *Causes:*
 - Defective or improper ramming of the sand.
 - Insufficient weighting of the mold during casting.
- *Remedies:*
 - This defect can be eliminated by ensuring proper weight over the molding boxes and ramming the sand properly.

Pouring Metal Defects

Misruns and cold shuts

- A misrun is a defect which lacks completeness of the casting due to the molten metal not filling the mold cavity completely.
- A cold shut is caused when two metal streams while meeting in the mould cavity, do not fuse together properly, thus causing a discontinuity or weak spot in the casting.
- A cold shut is an external defect formed due to improper fusion of two streams of molten metal poured in the cavity.

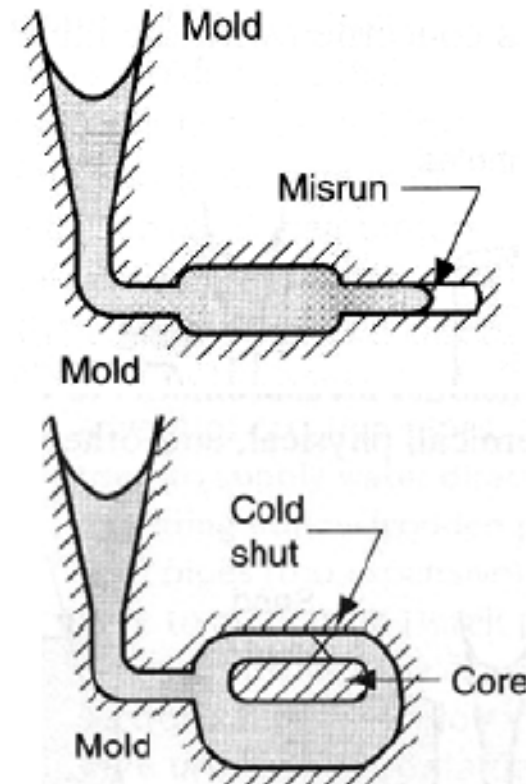
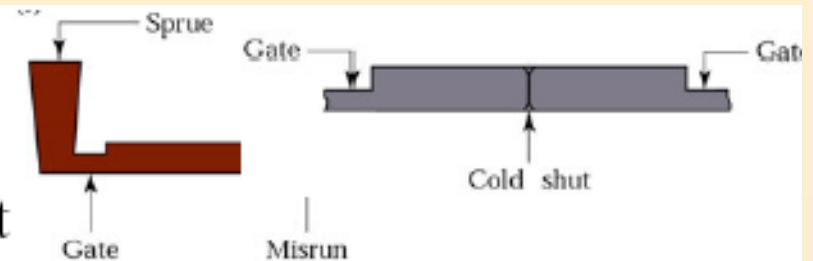


– *Causes:*

- Improper gating systems.
- Slow and intermittent pouring.
- Poor fluidity of metal.

– *Remedies:*

- To eliminate these defects, the gating system should be properly designed and the pouring should be controlled suitably.



Slag inclusions or Honey Combing:

- During the melting process, flux is added to remove the undesirable oxides and impurities present in the metal.
- It is an external defect consisting of number of small cavities in close proximity.
- *Causes:*
 - Suspended dirt in molten metal.
 - Incorrect gating and poor fluxing of metal.
- *Remedies:*
 - This defect can be avoided by preventing the slag from entering along with the molten metal.

Metallurgical Defects

Hot tears:

- Since metal has low strength at higher temperatures, any unwanted cooling stress may cause the rupture of the casting.
- It is an external or internal cracks occurring immediately after the metal have solidified, resulting from hindered contraction.
- *Causes:*
 - Abrupt changes in sections of the casting.
 - Poor collapsibility of the mold and core materials, which places extra stress on certain parts.
- *Remedies:*
 - This defect can be eliminated by ensuring proper casting design and using mold and core material having proper collapsibility.

